HONG KONG BAPTIST UNIVERSITY
SCHOOL OF CONTINUING EDUCATION

Higher Diploma in Data Science
Teaching and Learning Plan

Semester:	2nd Sem, AY 2025 - 26
Course Code and Title:	HDDS1103 Object-oriented Programming
Course Instructor:	Mr. Ben Fung
Course Duration:	13 weeks x 3 hours
Consultation Hours:	**Monday** afternoon (15:00 – 17:00, by appointment)
Email:	benfungcl@hkbu.edu.hk
QF Level:	4
Offering Department:	School of Continuing Education (SCE)
Prerequisite:	Introduction to Computing and Programming
Precursor/Exclusion:	Nil
Medium of Instruction:	English

Course Aims:

This course provides students with a deeper understanding of computer programming. Students will learn how to analyse and handle more difficult problems in computing and apply objected-oriented methodology in software development using Java. Topics include objects and classes, inheritance and polymorphism, exception handling, file handling and simple user interface design.

Course Intended Learning Outcomes (CILOs)

By the end of this course, students should be able to:
CILO 1: Describe the basic concepts of object-oriented programming;
CILO 2: Design applications and systems using object-oriented programming concepts and principles;
CILO 3: Develop solutions, programs and applications using Java programming language;
CILO 4: Integrate existing source code and libraries efficiently using advanced object-oriented concepts.

Teaching and Learning Activities (TLAs)

CILOs	TLAs will include the following:
CILOs 1–4	• Lectures with presentation slides, case studies and trend analysis on object-oriented programming languages and other related topics.
CILO 1	• Class discussion to deepen students' understanding of the features and advantages of using object-oriented programming languages to solve problems. • Examples using different programming languages to illustrate their similarities and differences.
CILO 2	• Class discussion on the pros and cons of system design from the object-oriented programming perspective. • Examples of system designs adopted in applications by enterprises and organizations.
CILO 3	• Demonstration of object-oriented based solutions to solve real life problems. • Practical exercises for students to develop programs using object-oriented concepts and principles.
CILO 4	• Practical exercises for students to evaluate and enhance existing programs and applications using object-oriented concepts.

Assessment Methods (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Participation	5%	1, 2, 3, 4	Students are expected to contribute actively to class discussions and activities.
Practical Laboratory Exercises	30%	1, 2, 3	Two individual laboratory exercises (15% each) will require students to design applications and systems by using object-oriented programming concepts.

Group Project	25%	1, 2, 3, 4	Students, in groups of 3 – 4, will be required to develop a Java program that provides specific functionalities to meet the requirements of a business case. The Java program source code should be submitted for assessment.
Individual Assignment	40%	2, 3, 4	Students will be asked to discuss possible goals for a given application system, identify challenges as well as develop and design solutions using appropriate Java platforms and tools. Students will be evaluated on their ability to integrate existing source code and libraries by applying object-oriented concepts.

Statement about Generative AI Tools use in Assessed Assignment Tasks

Given the proliferation of generative AI tools and their utility in teaching and learning, in alignment with the University's relevant principles, the default approach is that students are empowered and guided to use generative AI tools in assessed assignment tasks. Such use must align with the University's academic integrity guidelines for students while ensuring they are achieving the PILOs and CILOs.

The "Principles for the Use of Generative AI Tools in Teaching and Learning, and Assessment" ("Principles") guides us to use generative AI tools for better learning outcomes. You may access the "Principles" on BUniPort (U-Wide Policies & Info -> Policies & Guidelines -> Quality Assurance, Teaching & Learning).

In this course, generative AI tools can be used in the assignment tasks in the specific way mentioned:

- explain or clarify concepts
- planning and brainstorming on projects
- grammar checks for the write-up

<u>Assessment Brief</u>

Task name:	Participation (In-class)
Task description:	Active participation and enthusiastic contribution to all class activities.
Task weight:	5%
Time required:	N/A
Submission date:	On-going throughout the Semester

Task name:	Practical Laboratory Exercise 1 (In-class)
Task description:	Individual laboratory exercise (15% each) will require students to design applications and systems using object-oriented programming concepts.
Task weight:	15%
Time required:	1.5 hour
Submission date:	Session 4

Task name:	Group Project (Take Home)
Task description:	Students, in groups of 3 – 4, will be required to develop a Java program that provides specific functionalities to meet the requirements of a business case. The program source code should be submitted for assessment.
Task weight:	25%
Time required:	4 weeks
Submission date:	Session 8

Task name:	Practical Laboratory Exercises 2 (In-class)
Task description:	Individual laboratory exercises (15% each) will require students to design applications and systems using object-oriented programming concepts.
Task weight:	15%
Time required:	1.5 hour
Submission date:	Session 10

Task name:	Individual Assignment (Take Home)
Task description:	Students will be asked to discuss possible goals for a given application system, identify challenges as well as develop and design solutions using appropriate Java platforms and tools. Students will be evaluated on their ability to integrate existing

	source code and libraries by applying object-oriented concepts.
Task weight:	40%
Time required:	5 weeks
Submission date:	Session 13

Course Schedule (Tentative)

Week (Wed)	Content	CILOs	Assessments
1 (14-Jan)	Introduction to Object-oriented programming	1	
2 (21-Jan)	Data Types, Variables, basic operations	1, 2	Group List
3 (28-Jan)	Decision Making and Repetition	1, 2	
4 (04-Feb)	Object, Class, and Data Encapsulation	1-3	Lab 1
5 (11-Feb)	Constructor, method, and function	1-3	
6 (25-Feb)	Inheritance, Polymorphism	1-4	
7 (04-Mar)	Abstraction	1-4	
8 (11-Mar)	Exception handling and IO	2-4	*GP (13/3, Fri)*
9 (18-Mar)	Arrays and Collections	1-4	
10 (25-Mar)	Graphical User Interface (GUI) programming	1-3	Lab 2
11 (01-Apr)	Advanced GUI programming	1-3	
12 (08-Apr)	Design Patterns	1-3	
13 (15-Apr)	Adv. topics – Package, libraries & new features	2-4	IA

Textbooks: -

Balagurusamy, E. (2024). *Programming with Java (7th ed.).* McGraw-Hill Education.

Recommended Readings: -

Burd, B. A. (2021). *Beginning programming with Java for dummies (6th ed.)*. John Wiley & Sons.

Davis, A. L. (2020). *Modern programming made easy: Using Java, scala, groovy, and JavaScript (2nd ed.)*. Apress.

Liang, Y. D. (2021). *Introduction to Java programming and data structures (12th ed.)*. Pearson.

Liguori, R., & Liguori, P. (2017). *Java pocket guide: Instant help for Java programmers.* O'Reilly Media.

Somashekara, M. T., Guru, D. S., & Manjunatha, K. S. (2024). *Object Oriented Programming with Java (2nd ed.)*. PHI Learning Pvt. Ltd.

Tony Gaddis (2018). *Starting Out with Java: From Control Structures through Objects*. 6/E. Pearson ISBN: 978-1292110660